

Sea Ice, Polynyas, and Productivity: The Evolving Role of Polynyas in the Southern Ocean



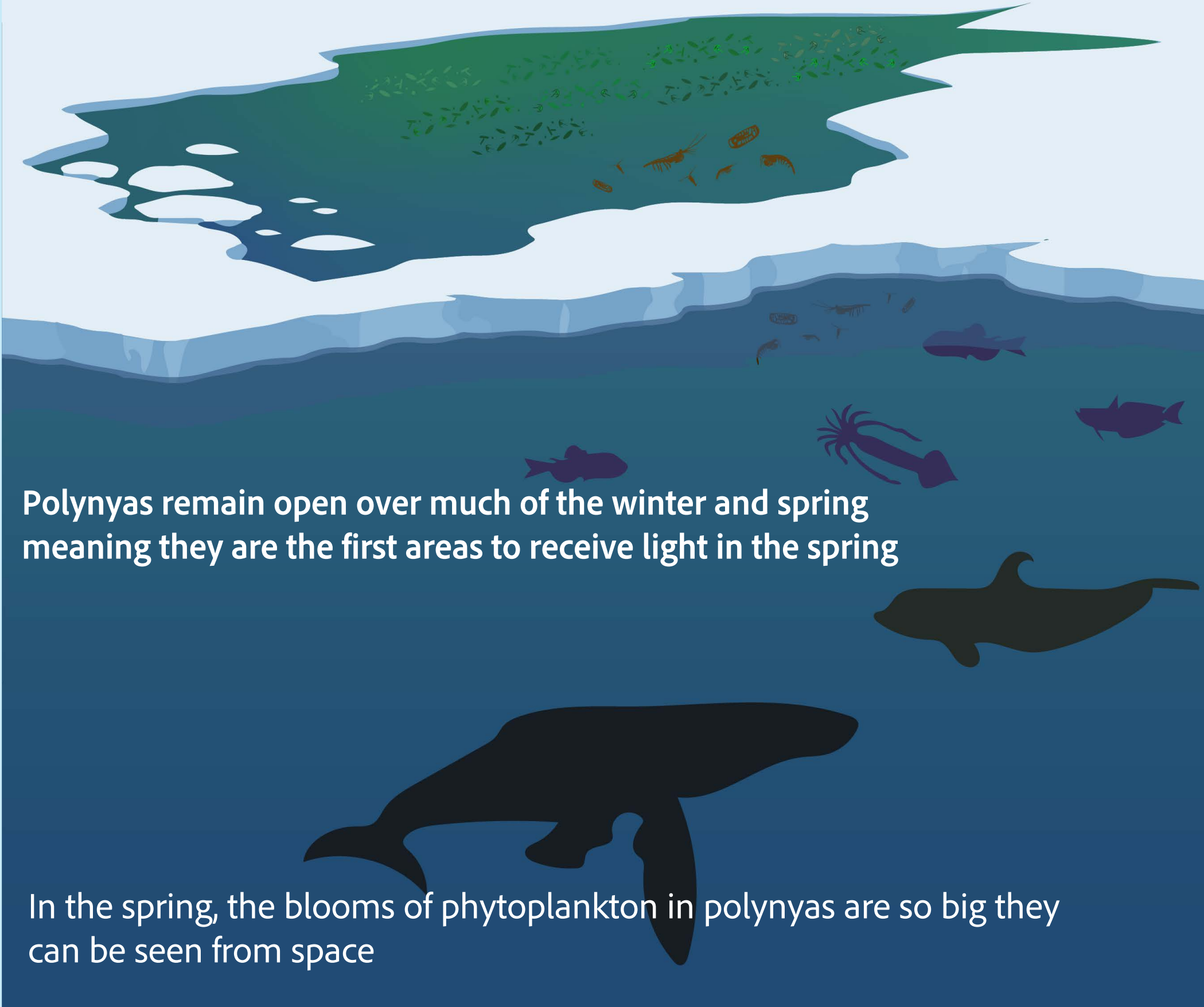
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MOTIVATION

Polynya

Origin: Russian meaning "hole in the ice"



Research Question:

How will the role of polynyas as biological hotspots change in the future?

APPROACH

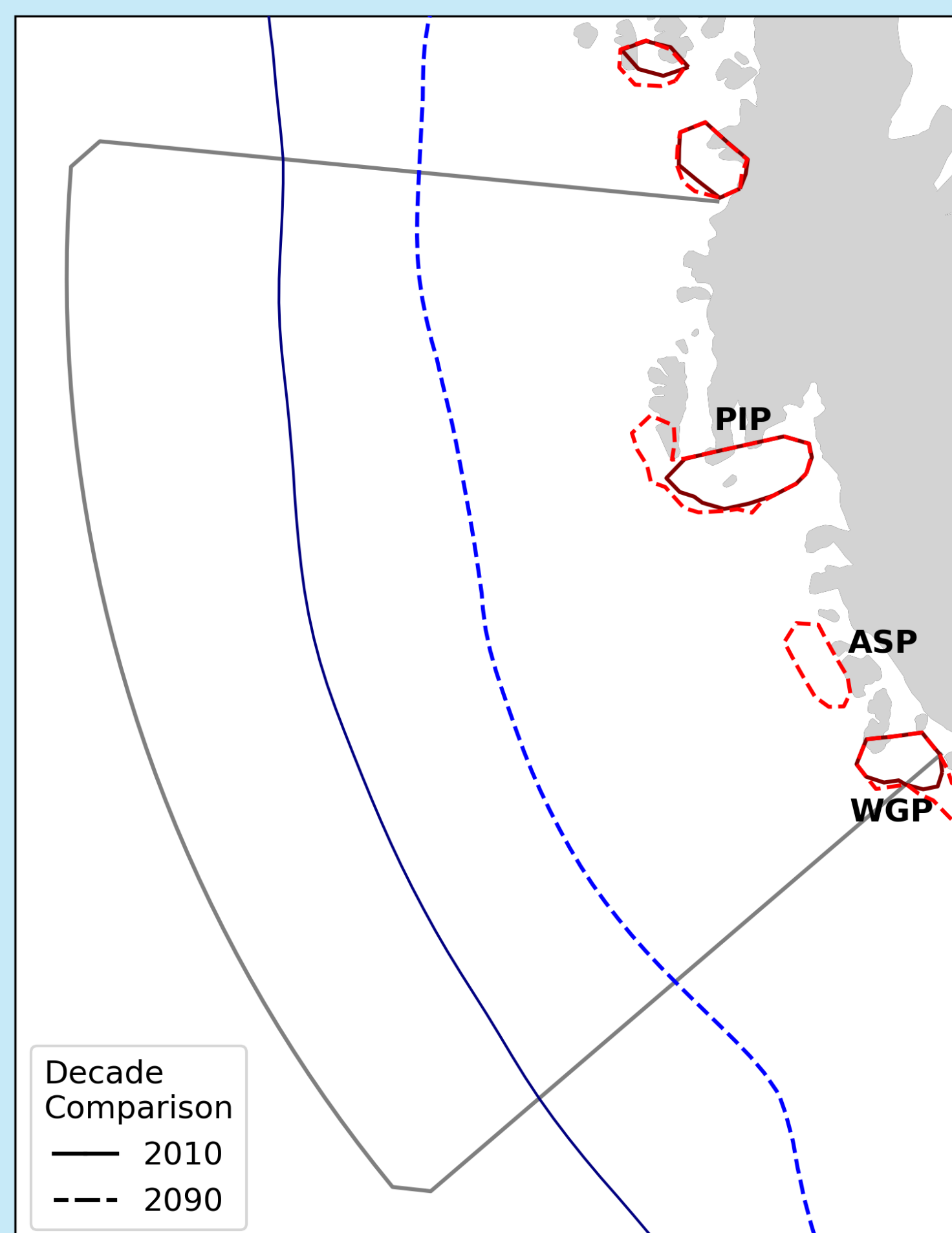
Community Earth System Model 2

- 5 Member Ensemble
- SSP3-7.0 forcing
- MARBL Ecosystem (4P2Z)
- 1° resolution

Regional Focus: Amundsen Sea

Dynamic system: rapid sea ice loss + opposing trends to west and east

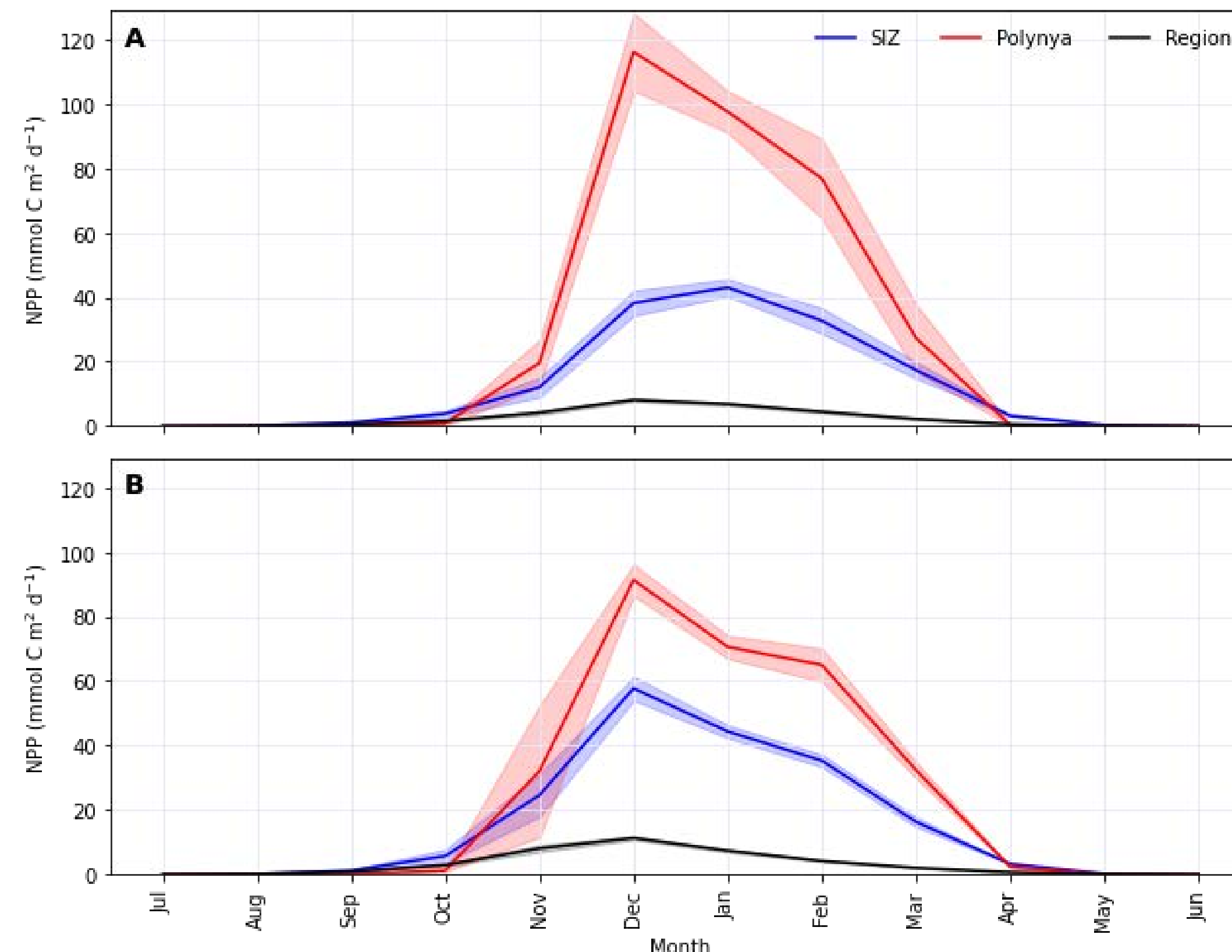
Projected favorable habitat for krill



PIP: Pine Island Polynya; ASP: Amundsen Sea Polynya

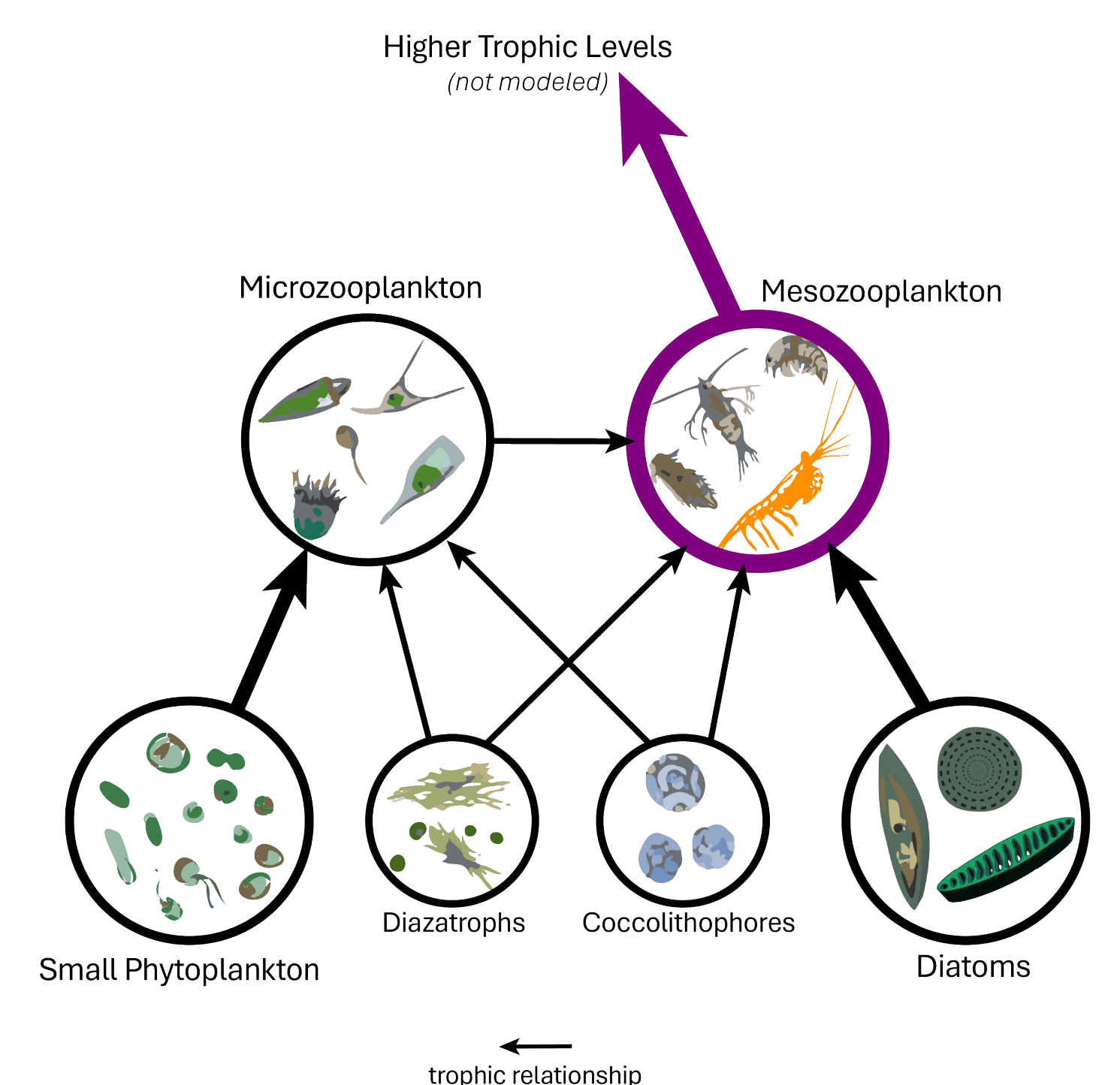
Polynyas: ≥10% detection in October
SIZ: mean winter IFRAC ≥15%

PROJECTED CHANGES



How could this shift in primary production impact mesozooplankton and higher-order consumers?

By the 2090s polynya NPP rates decline yet remain more productive than the SIZ until the winter



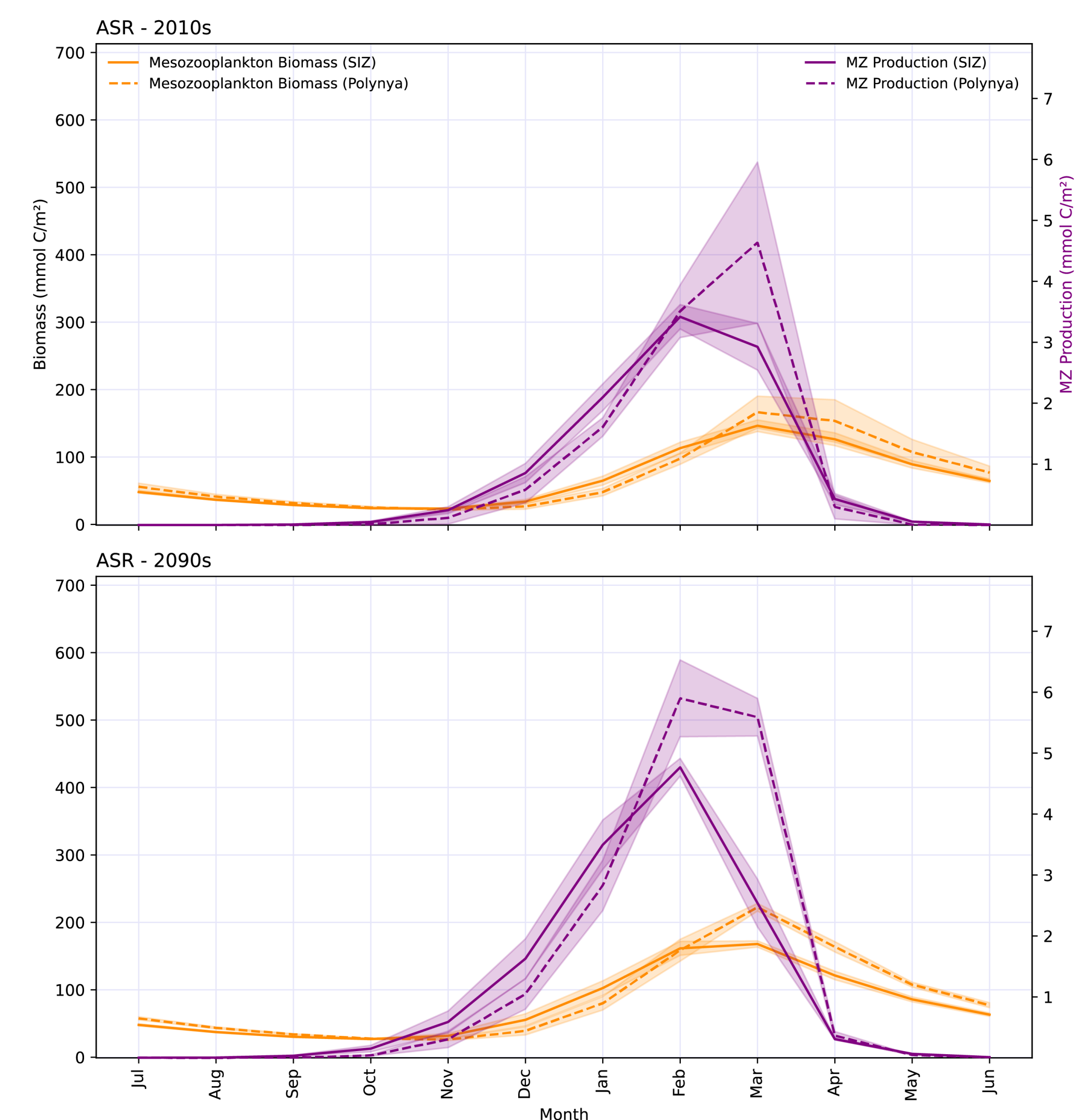
Key Findings

The NPP gap between polynyas and SIZ narrows, but the MZP gap widens

By the 2090s, seasonal MZP rates shifted: By February, MZP in polynya footprints outpaced SIZ MZP

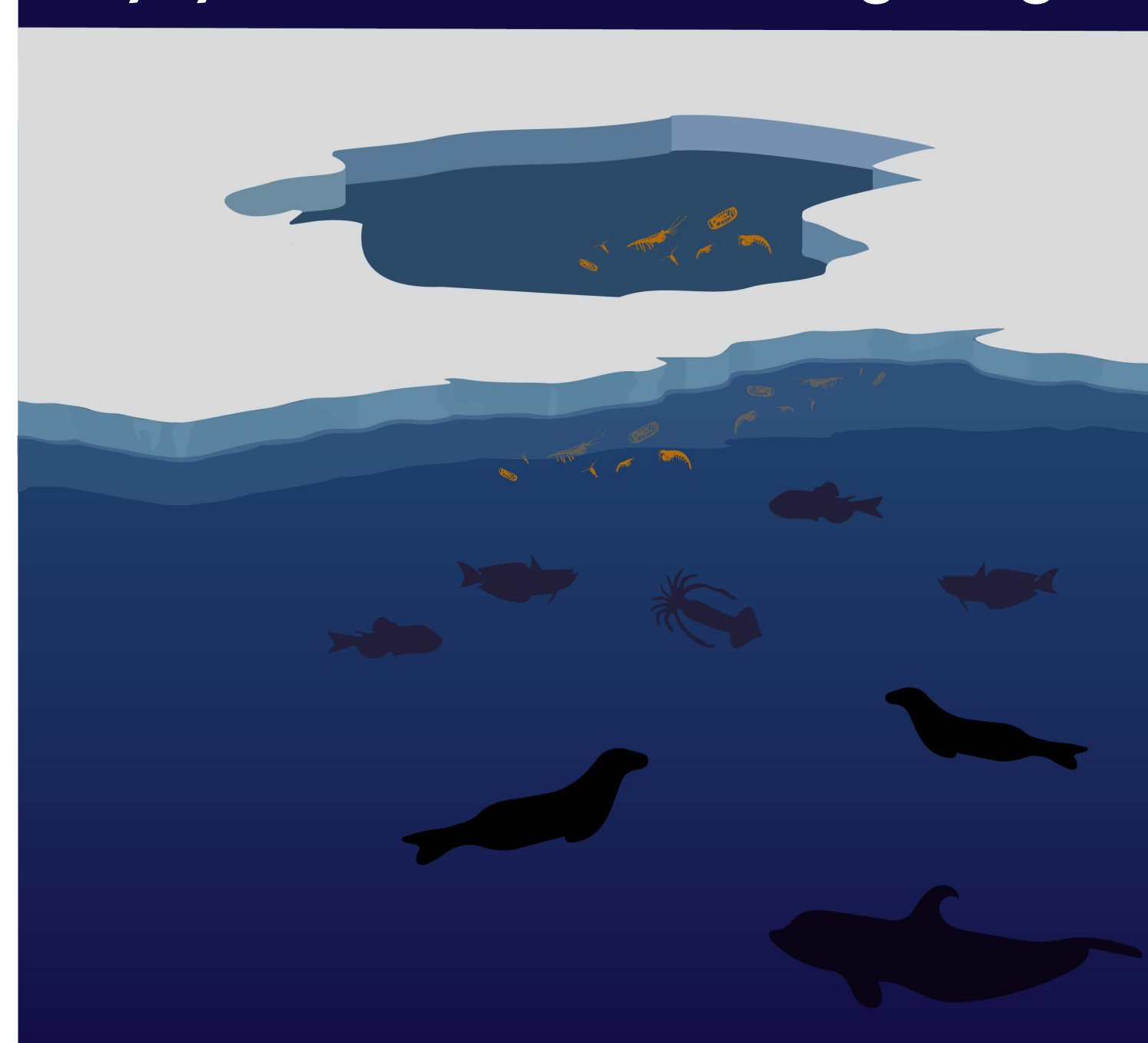
Where diatoms persist (polynyas), MZP is sustained. Where small phytoplankton dominate (SIZ), MZP is reduced.

The resulting increased mesozooplankton biomass is higher through the winter in polynya regions compared to the SIZ



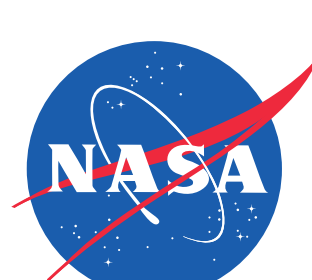
IMPLICATIONS AND FUTURE DIRECTIONS

Polynyas shift to overwintering refuge?



What these projections could mean:

- 1) Higher MZP biomass in winter = more food for krill and their predators (seals, penguins, whales).
- 2) In the Amundsen Sea, the ecological significance of polynya locations may increase even as their physical distinctiveness decreases.



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